

How Experience Shapes Future Learning

Research Statement | Continual Learning and Reinforcement Learning

As foundation models become more capable, we ask them to do more than solve isolated tasks. We expect them to keep learning as their environments change and to pursue longer projects, such as investigating a research question through a series of experiments. This requires them to retain useful knowledge, recognize when earlier lessons no longer apply, and make sense of new evidence. I am interested in how past experience shapes this continuing ability to learn.

My first-author manuscript, *Stability-Plasticity Theory for Continual Learning under Drifting Task Optima*, studies a linear model that learns tasks in sequence. Past prediction errors determine a diagonal metric that scales parameter updates and remains fixed during each task. The analysis establishes conditions for tracking changing task optima with bounded error. It also relates error on earlier tasks, measured relative to their optimal parameters, to current-task error and differences between task optima. In a three-task example, changing task order changes the learned metric and retention even though the pairwise distances between task optima are unchanged. Disabling metric learning removes this difference in that example.

In my work on automated research workflows, I observed agents repeatedly modifying the same approach despite little progress. This drew my attention to how agents use experience to decide what to try next. An unsuccessful replication may provide a reason to reconsider an earlier result; a failed experiment may help identify which assumption to test next.

These experiences have shaped two research interests: what to retain from experience to support further learning, and how agents learn to make decisions that put new evidence to use.

1. What should memory preserve for a changing learner?

As models continue to learn, the usefulness of stored experience can change. Self-adaptation methods evaluate training material through the current learner’s subsequent performance [1], and cross-model memory transfer studies how a new model can use memory built for another [2]. I am interested in how this dependence should affect what we preserve. A summary may record a useful rule while omitting cases the original model could not explain. A later learner may be able to explain those cases, provided the relevant evidence is still available.

Gradient-based memory methods provide a concrete starting point. K-priors reconstructs past gradients at future parameter values and considers changes in the model [3]. Gradient-matching coresets select small weighted subsets that approximate past-data gradients over a parameter distribution [4]. These results motivate a specific question: how well does the gradient error used to select a memory predict its value after the learner’s features change? A memory that closely matches the old model’s gradients may still miss patterns in its prediction errors that become learnable once new features are available.

A regression model with squared loss and features added from a fixed family offers a tractable way to examine this question. The task and original observations can stay fixed, while the learner gains access to patterns it previously could not represent. Gradient-matching memories, stored model predictions, and retained labeled examples provide useful comparisons. Relating their gradient errors to the loss on the original data after a fixed number of updates would connect the memory’s selection criterion to its later use. Learning from the full history provides a reference for measuring the cost of compression. The analytical target is how this gap depends on the retained evidence and the range of possible feature changes, with all auxiliary model state counted toward storage.

LESS offers an approach to inexpensive assessment: it estimates data influence and reports that selected data can transfer across model sizes and families [5]. Building on such methods, the aim is to understand when a cheap assessment can guide memory selection as a learner changes, and when additional evidence is needed. This would inform the choice between retaining original examples and using a more compact learned representation. Tests following controlled changes in a language model’s knowledge could examine whether the analysis helps explain both retention and the ability to correct earlier errors.

2. How can feedback help agents learn from experiments?

Repeating an experiment can lower our estimate of a configuration’s performance while helping us avoid a poor final choice. Bayesian selection formalizes this trade-off by weighing expected gains in the final choice against the cost of obtaining more evidence [6]. My interest is in how agents learn to make these judgments from experimental records, especially when feedback is easier to obtain for immediate score changes than for the value of further investigation.

Astar learns improvement feedback from historical experiments and refreshes its reward model through real execution [7]. Rewarding Progress analyzes when one policy supplies helpful process signals for another, linking improvement to how those signals distinguish and rank actions [8]. These results motivate examining a specific feedback rule in relation to both the final objective and its use in training. One example is the predicted probability that an experiment beats a reference score: under what conditions does using this probability to rank experiments favor the decisions that improve the final choice?

A small Bayesian selection problem provides an analytical benchmark. Suppose a finite set of configurations has unknown average performance on fresh data, and experiments produce noisy observations. The objective is the selected configuration’s expected performance minus experiment costs. With a tractable prior and observation model, finite observation outcomes, and a finite horizon, backward induction can compute the values of repeating an experiment, testing another configuration, and stopping. Comparing the resulting experiment rankings with rankings based on exact improvement probabilities separates limitations of the rule from prediction errors. A reward used throughout a trajectory requires a separate check of the objective it induces; accuracy on individual labels is not enough to settle that question.

The experimental record matters to this comparison. Results obtained on a shared validation split can contain different evidence from results obtained on independent splits, even when their reported scores agree. Work on approximate information states supplies a framework for asking which history summaries preserve the values needed for a decision [9]. In an explicit noise model, comparing summaries that retain or omit these dependencies could identify when feedback reverses the preferred action and which statistics prevent that error. Keeping the acting agent’s observations fixed would isolate the effect of information supplied to the feedback model.

Process feedback is one possible way to improve learning in this setting. RUDDER shows how delayed rewards can be redistributed while preserving return [10]. The practical question is whether a particular feedback construction improves learning over terminal rewards and direct decision-value learning for the same objective, including the cost of producing supervision. Controlled selection problems offer calculable reference decisions; longer research workflows can test whether the resulting explanation survives when the noise model and decision values must be estimated.

Research perspective

Across these directions, I am interested in how the way we represent experience affects what a learner can do with it later. My theoretical background leads me to start with settings where the relevant errors and assumptions can be examined directly. The goal is to develop explanations that guide concrete choices about memory and feedback, and to test those explanations as the learner or task becomes more complex.

References

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